



2022-2023 Term 2

March 19th, 2023; Time Allowed: 3 Hours

CUHKSZ ID

ZOOM session No.

- **Show all your work.** Correct answers with little supporting work will not be given credit.
- Closed Book Exam: One piece of double-sided A4 reference paper, a scientific calculator, and a paper-based dictionary are allowed.
- Unless approved by the instructors, students who arrive more than 30 minutes late will NOT be admitted.
- The total points are 120 points. You need to finish ALL the questions in 3 hours (180 minutes).

[illegible]

Summary of Basic Calculus:

$$\sin 2\theta = 2 \sin \theta \cos \theta,$$

$$\frac{d}{dx} x^n = nx^{n-1},$$

$$\frac{d}{dx} e^{ax} = ae^{ax},$$

$$\frac{d}{dx} \ln ax = \frac{1}{x},$$

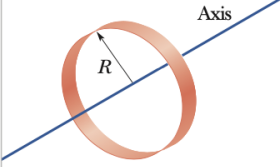
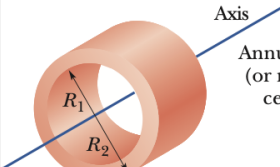
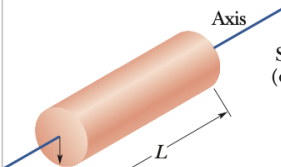
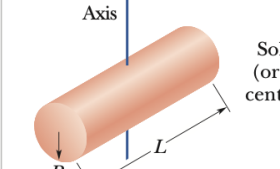
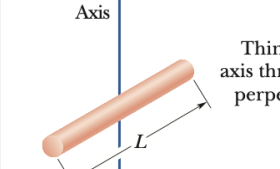
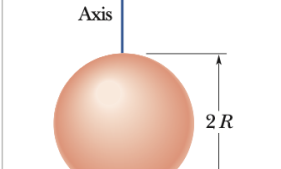
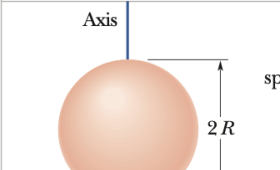
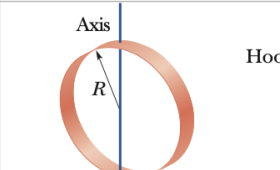
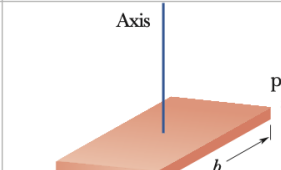
$$\frac{d}{dx} (uv) = v \frac{d}{dx} u + u \frac{d}{dx} v,$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad (n \neq -1),$$

$$\int \frac{dx}{x} = \ln x + C,$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad C \text{ is a constant.}$$

Table of Moments of Inertia:

 Hoop about central axis $I = MR^2$ (a)	 Annular cylinder (or ring) about central axis $I = \frac{1}{2}M(R_1^2 + R_2^2)$ (b)	 Solid cylinder (or disk) about central axis $I = \frac{1}{2}MR^2$ (c)
 Solid cylinder (or disk) about central diameter $I = \frac{1}{4}MR^2 + \frac{1}{12}ML^2$ (d)	 Thin rod about axis through center perpendicular to length $I = \frac{1}{12}ML^2$ (e)	 Solid sphere about any diameter $I = \frac{2}{5}MR^2$ (f)
 Thin spherical shell about any diameter $I = \frac{2}{3}MR^2$ (g)	 Hoop about any diameter $I = \frac{1}{2}MR^2$ (h)	 Slab about perpendicular axis through center $I = \frac{1}{12}M(a^2 + b^2)$ (i)

1. Given the two vectors

$$\vec{a} = 4\hat{i} + 3\hat{j} + 5\hat{k}, \quad \text{and} \quad \vec{b} = -6\hat{i} + 3\hat{j} - 4\hat{k},$$

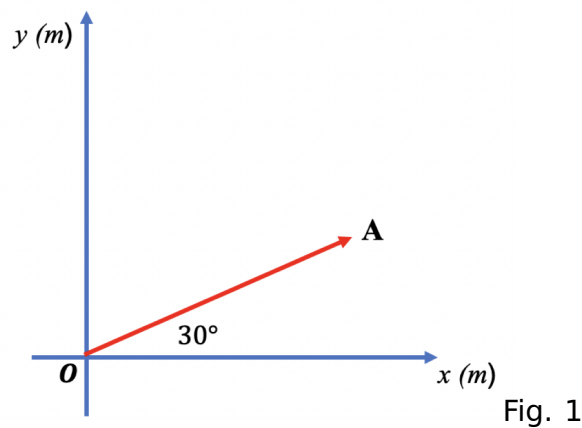
answer the following questions. (All the numbers are exact in this problem. No need to worry about the significant figures.) **(10 pts)**

(a) Find $\vec{a} + \vec{b}$. **(2 pts)**

(b) Find the vector product $\vec{a} \times (3\vec{b})$. **(4 pts)**

(c) Find the scalar product of the above two vectors $(\vec{a} + \vec{b}) \cdot (\vec{a} \times 3\vec{b})$. **(4 pts)**

2. An object of 2.00 kg moves from point O with initial velocity of 5.00 m/s along positive y direction. After 2.00 seconds, it reaches point A , which is 20.0 m away from point O as shown below. Suppose that there is only one non-zero constant force acting on this object. **(10 pts)**



- (a) What kind of motion does this object do? (Hint: constant velocity, constant acceleration, rotation, free fall or any other motions.) **(2 pts)**
- (b) Find the force acting on the object. Please indicate the magnitude and direction of the force. **(4 pts)**
- (c) Draw (sketch) the trajectory of the object in the above figure, and find the trajectory function relation between x and y . **(4 pts)**

3. As shown in Fig. 2, an inventor designs a pendulum clock using a bob with mass 10.0 kg at the end of a thin wire of length $L = 1.00\text{ m}$. Instead of swing back and forth, the bob is to move in a horizontal circle with constant speed v with the wire making a fixed angle $\beta = 30.0^\circ$ with the vertical direction. This is called a conical pendulum because the suspending wire traces out a cone. **(10 pts)**

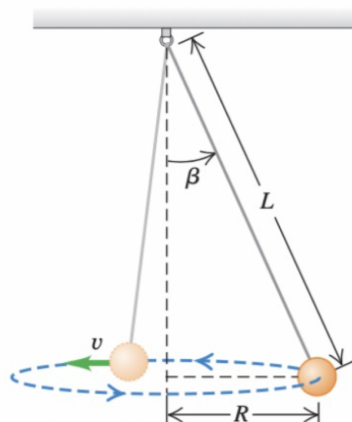


Fig. 2

- (a) Find the speed v of the bob. **(4 pts)**
- (b) Find the period T , which is the time needed for the bob to complete a circle. **(2 pts)**
- (c) If a new bob with mass 20.0 kg is used to replace the original one, will the period be affected? **(4 pts)**

4. Only one force is acting on a 2.0 kg particle-like object whose velocity is given by **(10 pts)**

$$v = 5.0 \text{ m/s} + (4.0 \text{ m/s}^2)t - (3.0 \text{ m/s}^3)t^2,$$

with v in m/s and t in seconds.

- (a) What is the displacement from $t = 0 \text{ s}$ to $t = 1.0 \text{ s}$? **(2 pts)**

- (b) What is the kinetic energy of this object at $t = 1.0 \text{ s}$? **(2 pts)**

- (c) What is the work done by the force from $t = 0 \text{ s}$ to $t = 1.0 \text{ s}$? **(2 pts)**

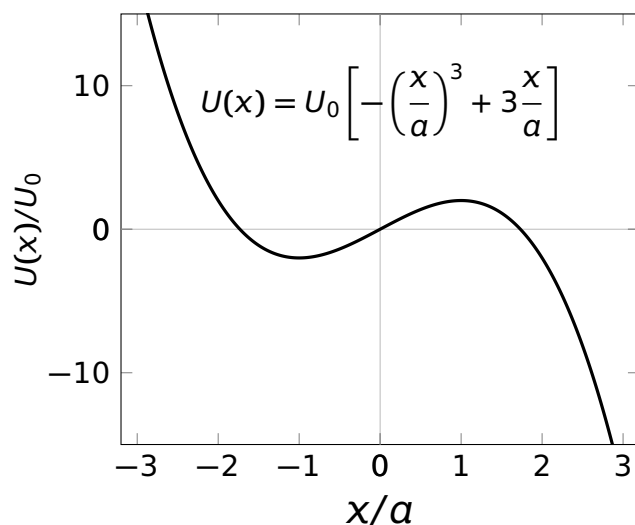
- (d) What is the instantaneous acceleration of this object at $t = 1.0 \text{ s}$? **(2 pts)**

- (e) What is the magnitude of the force at $t = 1.0 \text{ s}$? **(2 pts)**

5. A force $F(x)$ acted on a particle is represented by the potential-energy function **(10 pts)**

$$U(x) = U_0 \left[-\left(\frac{x}{a}\right)^3 + 3\frac{x}{a} \right],$$

where a is constant with the dimension of length. $U(x)$ is plotted in the figure below.



- (a) Find the force $F(x) = -\frac{dU(x)}{dx}$ from the potential given above. **(2 pts)**

- (b) At what value of x is the force $F(x)$ zero? **(3 pts)**

- (c) The location where the force equals zero is called the equilibrium of this potential. Are these equilibrium positions stable or unstable? Explain why. **(5 pts)**

6. A wedge of mass M is initially at rest on a frictionless and horizontal surface. A block of mass m is moving on the horizontal surface toward the wedge with an initial velocity v_0 . The block can move upward along the wedge, which also has a frictionless surface, until the block's center of mass reaches the highest position with a height h relative to its initial position (see figure below). **(10 pts)**

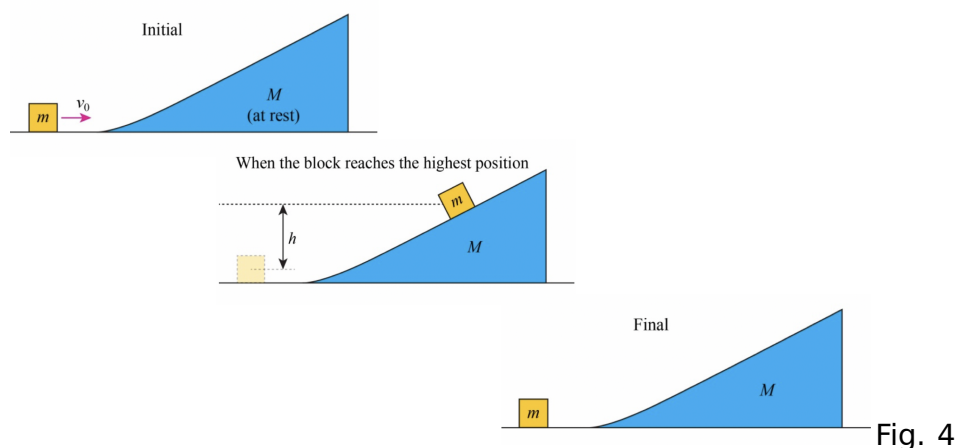


Fig. 4

- (a) What are the speeds of the block and the wedge when the block reaches the highest position? **(2 pts)**
- (b) What is the height h for the highest position of the block? **(3 pts)**
- (c) Later on, the block moves downward along the wedge and separates with the wedge at the end. What are the final speeds of the block and of the wedge? **(4 pts)**
- (d) Check your calculation plausibility by considering the limiting case when $M \gg m$. **(1 pts)**

7. A water storage truck is moving forward at $v = 3.6$ km/h along a straight and frictionless street when its driver decides to spray water. Water is sprayed at a constant rate of 2.0 kg/s at a speed (relative to the truck) of $v_s = 2.5$ m/s. The initial mass of the truck with water is $M = 2.0 \times 10^3$ kg. (Hint: This is a mass-changing problem.) **(10 pts)**

(a) If the truck accelerates forward when the drive starts spraying water, does he spray water forward or backward (relative to the truck)? Why? **(2 pts)**

(b) If the truck sprays water forward, what is the force applied to the truck? Specify the magnitude and direction. **(3 pts)**

(c) If the truck sprays water backward, what is the instantaneous acceleration of the truck at the moment when it just starts spraying water? **(2 pts)**

(d) If the truck keeps spraying water backward for a while, find the time-dependent acceleration of the truck. (Hint: find the time dependence of the truck's mass $M(t)$ first.) **(3 pts)**

8. Rotating Wheel: A wheel with radius r is lying in the plane of your paper and rotating counter-clockwise with angular velocity $\vec{\omega}$. Choose a point P on the rim and draw a vector $\vec{r}$ from the center of the wheel to that point. **(10 pts)**

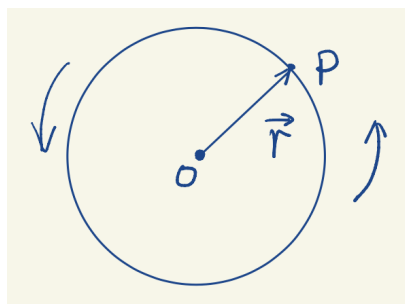


Fig. 5

- (a) What is the direction of angular velocity $\vec{\omega}$? (Upward, perpendicular to the paper or Downward, Perpendicular to the paper) **(2 pts)**
- (b) Find both the direction (indicate or describe) and magnitude of the velocity $\vec{v}$ at point P. **(2 pts)**
- (c) Suppose ω is a constant, find the magnitude and direction of the linear acceleration for the point P point on the wheel. **(2 pts)**
- (d) Suppose the angular acceleration α is in the same direction of ω , which means the angular velocity increases, find the magnitude and direction of the total acceleration for the point P on the wheel. (Draw a diagram to show the direction.) **(4 pts)**

9. Fig. 6 shows an early method of measuring the speed of light that makes use of a rotating slotted wheel. A beam of light passes through one of the slots at the outside edge of the wheel, travels to a distant mirror, and returns to the wheel just in time to pass through the next slot in the wheel. One such slotted wheel has a radius of 5.00 cm and 500 slots around its edge. Measurements taken when the mirror is $L = 500$ m from the wheel indicate a speed of light of 3.00×10^8 m/s. **(10 pts)**

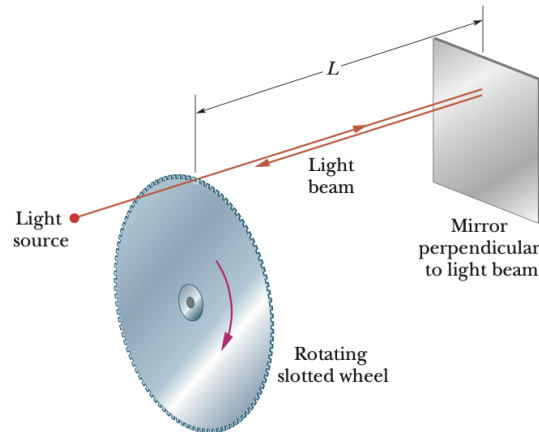


Fig. 6

- (a) How much time does the light take to go from the wheel to the mirror and back again? **(2 pts)**
- (b) What is the angle between two slots next to each other? **(2 pts)**
- (c) What is the angular speed (constant) of the wheel? **(3 pts)**
- (d) What is the linear speed of a point on the edge of the wheel? **(3 pts)**

10. Moments of Inertia (Rotational Inertia) of thin rods.

(10 pts)

- (a) Demonstrate that the moment of inertia of a thin uniform rod is $I_{cm} = \frac{1}{12}ML^2$ when it is rotating about the axis through the center, perpendicular to the rod. **(2 pts)**

- (b) Find the moment of inertia of a thin uniform rod of length L and mass M when it is rotating about its left end, perpendicular to rod, as shown below. **(2 pts)**

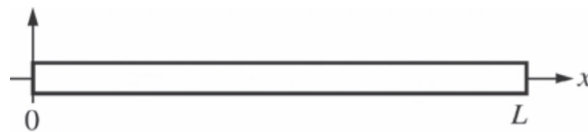


Fig. 7

- (c) Now consider a non-uniform rod with the linear density σx where x is the distance from the origin shown in part (b). Given the length L and mass M for this rod, find σ in terms L and M . **(3 pts)**

- (d) Find the moment of inertia of the non-uniform rod described in part (c), when it is rotating about its left end, perpendicular to rod. **(3 pts)**

11. Figure below shows a disk of mass M and radius R is held up by a massless string. **(10 pts)**

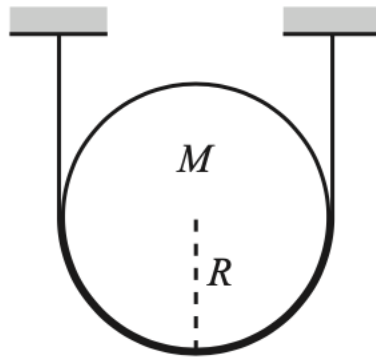


Fig. 8

- (a) Suppose the surface of the disk is frictionless. What is the tension T in the string supporting the disk? **(2 pts)**
- (b) Following part (a), is the magnitude of the tension along the string the same everywhere? YES/NO. **(1 pts)**
- (c) Let there now be friction between the disk and the string, with coefficient μ . Is the magnitude of the tension along the string the same everywhere? YES/NO. **(1 pts)**
- (d) Consider the friction between the disk and the string, with coefficient μ . What is the smallest possible tension in the string at its lowest point? **(5 pts)**
- (e) Consistency check: What is the minimum tension $T_{\min}$ when $\mu \rightarrow \infty$? **(1 pts)**

12. Moment of inertia (rotational inertia) of the regular polygon:

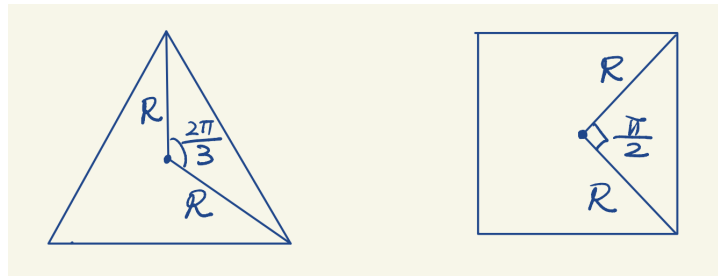


Fig. 9

- (a) Find the moment of inertia of a regular triangle (as shown in above left figure) plate (with radius R and mass M) about an axis through its center perpendicular to the plate. (Hint: You may find the parallel axis theorem useful in solving this problem.) **(2 pts)**

- (b) Find the rotational inertia of a square plate with radius R and mass M as shown in above right figure) about an axis through its center perpendicular to the plate. (Hint: You may find the table on Page 1 useful and you can use the result without derivations.) **(2 pts)**

- (c) Find the rotational inertia of a regular polygon¹ with radius R and mass M about an axis through its center perpendicular to the plate. **(5 pts)**

- (d) What is the corresponding moment of inertia when one takes $N \rightarrow \infty$? (Hint: what does a regular polygon look like when $N \rightarrow \infty$?) **(1 pts)**

¹A regular polygon is a polygon that is direct equiangular (all angles are equal) and equilateral (all sides have the same length).